

Types of Stretching

Stretching techniques typically fall into three broad categories: static, dynamic, and precontraction (73).

Static Stretching

The static stretching approach requires the movement of a body segment to a point in the ROM where resistance to further movement is experienced (i.e., the end position). This stretching action can be performed either actively or passively. An **active stretch** occurs when the person stretching supplies the necessary forces to enable the end position to be reached. During the sitting toe touch, for example, the athlete contracts the abdominal muscles and hip flexors to flex the torso forward to stretch the hamstrings and low back. A **passive stretch** occurs when a partner or a stretching machine provides external force to cause or enhance a stretch.

A **static stretch** involves a slow and constant movement toward the end position, which is then held for 10 to 30 seconds (66). This duration is thought to cause an acute viscoelastic relaxation response, facilitating relaxation and concurrent elongation of the stretched muscle (34). Because it is performed slowly, static stretching does not elicit the stretch reflex of the stretched muscle (71); therefore, the likelihood of injury is less than during ballistic stretching (71). In addition, static stretching is easy to learn and has been shown to effectively improve ROM (38). Although injury to muscles or connective tissue may result if the static stretch is too intense, there are no real disadvantages to static stretching as long as proper technique is used. Static stretching is appropriate for all athletes in a variety of sports for increasing flexibility.

The sitting toe touch is a good example of a static stretch. To perform this stretch statically, the athlete sits on the ground with the lower extremities together and knees extended, leans forward from the waist, and slowly reaches toward the ankles. The athlete

gradually increases the intensity of the stretch by leaning forward until he or she feels mild discomfort in the hamstrings or lower back. The athlete holds this position for 10 to 30 seconds and then slowly returns to an upright sitting position. The stretch is static because it is performed slowly and the end position is held without movement.

Dynamic Stretching

Dynamic stretching is a broad term used to describe actions that involve active motions where a performer moves actively in a controlled and moderate paced manner through a typically large ROM (66). Often this approach is used as a type of functionally based stretching exercise that uses sport-generic and sport-specific movements to prepare the body for activity (48). Dynamic stretching—sometimes referred to as **mobility drills** (3)—places an emphasis on the movement requirements of the sport or activity rather than on individual muscles. This type of exercise can closely duplicate the movement requirements of a sport or activity (48); for example, a walking knee lift stretch mimics the knee lift of a sprinter.

The ability to actively move a joint through a ROM is generally far more sport specific than the ability to statically hold a stretch. Advantages of dynamic stretching include the ability to promote mobility and to replicate the movement patterns and ROM required for sport activities. As a consequence, dynamic stretches, as outlined in the discussion of warm-up, are increasingly the preferred method of increasing the ROM during warm-up.

In dynamic stretching, unlike static stretching, the muscle does not relax during the stretch but instead is active through the ROM; this is also more specific to the movement patterns that occur in sport. Even though it is an ideal warm-up activity, dynamic stretching may be less effective than static or proprioceptive neuromuscular facilitation stretching at increasing static ROM (71); in situations in which an increased static ROM is needed, static or proprioceptive neuromuscular facilitation methods may be preferred.

When one is designing a dynamic stretching program, the starting point should be a careful analysis of the major movement patterns within the given sport as well as an analysis of fundamental movement patterns that can facilitate performance and the ROM required within these movements. One can then select exercises that replicate those movements via a series of dynamic stretches. In this way, it is possible to achieve a highly specific flexibility program.

Dynamic stretching also provides the opportunity to combine movements (48). This gives the strength and conditioning professional a large number of combinations that can be used to provide variety in the warm-up. Athletes can perform dynamic stretching exercises either for a series of repetitions in the same place (e.g., 10 lunges) or for a series of repetitions to cover a given distance (e.g., lunge for 15 m [16 yd]). Regardless of the method chosen, each drill should start at an appropriate ROM and gradually increase this during subsequent repetitions or sets. Additionally, the speed of movement can also be increased but always under a high level of control. For example, athletes can perform the knee lift exercise over a distance of 15 meters, starting at a walk and building to a skip in subsequent repetitions.

In a dynamic stretch that mirrors a sport skill—such as a sprinter's knee lift drill—it is important that the stretch also emphasize the key skill factors required for the movement so that the most important mechanics of the drill are reinforced. For example, if the knee lift drill is used in the warm-up, effective body mechanics should be emphasized along with key joint positions such as dorsiflexion of the ankle of the lifted foot. The use of dynamic stretches must always be coordinated with appropriate sport techniques and never compromise proper technique.

A **ballistic stretch** can be considered a special type of dynamic stretch. Ballistic stretches similarly involve active muscular efforts, but rather than performing these under control, they instead use intense bouncing-type movements to achieve the required ROM (66). Ballistic stretching, if not appropriately controlled or

sequenced, may injure muscles or connective tissues, especially when there has been a previous injury (20). Ballistic stretching usually triggers the stretch reflex that does not allow the involved muscles to relax, which may limit ROM, so the strength and conditioning professional should monitor this during the session.

For example, consider the sitting toe touch performed as a ballistic stretch rather than a static stretch. The athlete sits on the ground with knees extended, lower extremities together, and upper body perpendicular to the legs. The athlete reaches quickly toward the ankles, bounces at the end position, and immediately returns to a near-vertical upper body position. With each repetition, the end position extends farther than in the preceding repetition. Ballistic stretching has been shown to be as effective as static stretching in enhancing ROM (58) and could play a role in a flexibility development program (22). However, care must be taken to ensure that the athlete is appropriately prepared for this type of exercise acutely and chronically, and special care should be taken when a previous injury has been reported.

Dynamic and ballistic stretches may appear similar; however, a number of key differences significantly alter the effects of these activities such that dynamic stretching avoids the potential negative effects associated with ballistic stretching. Dynamic stretching avoids bouncing and is performed in a more controlled manner than ballistic stretching. The result is a controlled ROM that is often smaller than that produced by ballistic stretching but demonstrates the control required to be able to actively move through the full ROM and assume and hold the end position.

Precontraction Stretching

Proprioceptive neuromuscular facilitation (PNF) is the most common precontraction stretching methodology. PNF was originally developed in the late 1940s as part of a neuromuscular rehabilitation program designed to relax muscles with increased tone or activity (1, 81). It has since been expanded to athletics as a method of

increasing flexibility. PNF techniques are usually performed with a partner and involve both passive movement and active (concentric and isometric) muscle actions. PNF stretching has been purported as being superior to other stretching methods for increasing flexibility (81), possibly because it activates the sensory receptors in the agonist and antagonist muscles (56), although evidence for this has not been consistently shown (38, 56). Additionally, PNF stretching can often be impractical because most of the stretches require a partner and some expertise.

During a PNF stretch, three specific muscle actions are used to facilitate the passive stretch. Both isometric and concentric muscle actions of the antagonist (i.e., the muscle being stretched) are used before a passive stretch of the antagonist to achieve autogenic inhibition (see the [Proprioceptor: Golgi Tendon Organs](#) sidebar on this page). The isometric muscle action is referred to as *hold* and the concentric muscle action as *contract*. A concentric muscle action of the agonist, called **agonist contraction**, is used during a passive stretch of the antagonist to achieve reciprocal inhibition. Each technique also involves passive, static stretches that are referred to as *relax*. When using static contractions, these should be held for approximately 3 to 6 seconds at no more than 20% of maximum voluntary contraction, with the first repetition generally giving the greatest benefit. One or two sessions per week are recommended (81).

Proprioceptor: Golgi Tendon Organs

The **Golgi tendon organs** (GTOs) are sensitive to increases in muscular tension. When stimulated, the GTO causes a muscle to reflexively relax. Relaxation that occurs in the same muscle that is experiencing increased tension is called **autogenic inhibition** (18, 22). Autogenic inhibition is accomplished via active contraction of a muscle immediately before a passive stretch of that same muscle. Tension built up during the active contraction stimulates the GTO, causing a reflexive relaxation of the muscle during the subsequent passive stretch. Relaxation that occurs in the muscle opposing the muscle experiencing the increased tension is called **reciprocal inhibition** (18). This occurs when one

simultaneously contracts the muscle opposing the muscle that is being passively stretched. Here the tension in the contracting muscle stimulates the GTO and causes a simultaneous reflexive relaxation of the stretched muscle.

There are three basic PNF stretching techniques: hold-relax (15, 18), contract-relax (15), and hold-relax with agonist contraction (18). The techniques are completed in three phases. With each of the three techniques, the first phase incorporates a passive prestretch of 10 seconds. The muscle actions used in the second and third phases differ for the three techniques; the second and third phases give each technique its name.

The following sections use the hamstring stretch to describe the three techniques; [figures 15.1](#) and [15.2](#) show the starting positions for the athlete and the partner.



FIGURE 15.1 Starting position of PNF hamstring stretch.



FIGURE 15.2 Partner and athlete leg and hand positions for PNF hamstring stretch.

Hold-Relax

The **hold-relax** technique begins with a passive prestretch that is held at the point of mild discomfort for 10 seconds ([figure 15.3](#)). The partner then applies a hip flexion force and instructs the athlete, “Hold and do not let me move the leg.” The athlete holds and resists the movement so that an isometric muscle action occurs and is held for 6 seconds ([figure 15.4](#)). The athlete then relaxes, and a passive stretch is performed and held for 30 seconds ([figure 15.5](#)). The final stretch should be of greater magnitude due to autogenic inhibition (i.e., activation of the hamstrings).



FIGURE 15.3 Passive prestretch of hamstrings during hold-relax PNF hamstring stretch.



FIGURE 15.4 Isometric action during hold-relax PNF hamstring stretch.



FIGURE 15.5 Increased ROM during passive stretch of hold-relax PNF hamstring stretch.

Contract-Relax

The **contract-relax** technique also begins with a passive prestretch of the hamstrings that is held at the point of mild discomfort for 10 seconds ([figure 15.6](#)). The athlete then extends the hip against resistance from the partner so that a concentric muscle action through the full ROM occurs ([figure 15.7](#)). The athlete then relaxes, and a passive hip flexion stretch is applied and held for 30 seconds ([figure 15.8](#)) while the athlete concentrically contracts the agonist. The increased ROM is facilitated due to autogenic inhibition (i.e., activation of the hamstrings). In an alternative to this technique, the athlete attempts to extend the hip and the partner does not allow the movement (63). Because this is essentially the same as the hold-relax technique, the contract-relax method described here is preferred.

Hold-Relax With Agonist Contraction

The **hold-relax with agonist contraction** technique is identical to hold-relax in the first two phases ([figures 15.9](#) and [15.10](#)). During the third phase, a partner applies force to the antagonist muscle groups while the athlete is required to maintain the position through a concentric action of the agonist, thus adding to the stretch force ([figure 15.11](#)). Following the isometric hold, the athlete flexes the

hip, thereby moving further into the new ROM. With this technique, the final stretch should be greater, primarily because of reciprocal inhibition (i.e., activation of the hip flexors) (63) and secondarily because of autogenic inhibition (i.e., activation of the hamstrings) (63).

The hold-relax with agonist contraction is the most effective PNF stretching technique due to facilitation via both reciprocal and autogenic inhibition.

See [pages 407](#) through [409](#) for descriptions and photos of PNF stretching exercises using a partner for the calves, chest, groin, quadriceps and hip flexors, and shoulders.



FIGURE 15.6 Passive prestretch of hamstrings during contract-relax PNF stretch.



FIGURE 15.7 Concentric action of hip extensors during contract-relax PNF stretch.



FIGURE 15.8 Increased ROM during passive stretch of contract-relax PNF stretch.



FIGURE 15.9 Passive prestretch during hold-relax with agonist contraction PNF hamstring stretch.



FIGURE 15.10 Isometric action of hamstrings during hold-relax with agonist contraction PNF hamstring stretch.



FIGURE 15.11 Concentric contraction of quadriceps during hold-relax with agonist contraction PNF hamstring stretch, creating increased ROM during passive stretch.

Programming Considerations for Stretching

As with all forms of training, frequency, duration, and intensity are important issues in program design (26). Both static and precontraction stretching have been shown to increase joint flexibility around the knee, hip, trunk, shoulder, and ankle joints (56, 89), although doubt remains as to which protocols elicit the greatest effects (56). This is further complicated by the fact that the exact mechanisms responsible for increased flexibility are still unclear. The acute effects of stretching on ROM are transient and are greatest immediately after the stretching session; then they decline, with the duration of significant improvements in flexibility ranging from 3 minutes (27) to 24 hours (29).

For longer-lasting effects, a dedicated flexibility program is required (89), and improvements in flexibility have been shown with short-term interventions ranging from two to eight weeks (38). Interestingly, time stretching per week seems to have a greater impact than time spent stretching within a single session, suggesting that frequency rather than duration may be a more important factor (90). It is important to note that PNF stretching may be an exception to this, where a single session per week may be optimal (66). In terms of intensity of stretching, higher intensities do not appear to have any additional benefit over less intense methods (54). However, it is not clear whether this applies when extreme flexibility is needed (54).

As to the appropriate duration for a static stretch, 15 to 30 seconds is generally recommended (81) and has been shown to be more effective than shorter durations (66). Evidence supports the use of 30 seconds (4-6) and suggests diminishing returns beyond this duration. Another consideration is that the total stretch time throughout a day may be as important as the duration of a single stretch (17), further emphasizing the potential importance of

frequency as a key factor in program design. When performing static stretches, athletes should hold the stretch at a position of mild discomfort (not pain); joint integrity should never be compromised to increase ROM. All stretching sessions should be preceded by a period of general activity to raise muscle temperature. Because neural and vascular structures are stretched during flexibility exercises, athletes should be monitored for a loss of sensation or radiating pain.

As a general guideline, stretching at the following times can provide optimal benefits:

- *Following practice and competition.* Postpractice stretching potentially facilitates ROM improvements (40), although the evidence for this is mixed (79). It should be performed within 5 to 10 minutes after practice. The increased body temperature increases the elastic properties of collagen within muscles and tendons, which potentially allows for a greater stretch magnitude. Postpractice stretching may also decrease muscle soreness, although the evidence on this is again ambiguous (59, 79).
- *As a separate session.* If increased levels of flexibility are required, additional stretching sessions may be needed. In this case, stretching should be preceded by a thorough general warm-up to allow for the increase in muscle temperature necessary for effective stretching. This type of session can be especially useful as a recovery session on the day after a competition.

Conclusion

A warm-up can provide benefits that enhance subsequent performance. Warm-up should be geared toward the particular sport or activity and should use an appropriate structure, ensuring that an athlete is optimally prepared for subsequent activity. Effective warm-up planning can ensure that the activities prepare the athlete for the upcoming session but do not induce undue fatigue. Additionally, the warm-up should be planned so that the activities contribute both to the goal of the upcoming session and to the development of the athlete in the medium and long term.

Optimal flexibility for performance varies from sport to sport and is closely related to the types of movements and actions an athlete will be required to perform. The concept of mobility may be more appropriate than flexibility with its focus on active movement through the required ROM. For athletes who need to increase flexibility, static and PNF stretching techniques can facilitate an effective increase in ROM, and these techniques should be a key component of an extended training program. Strength and conditioning professionals should consider each athlete's unique combination of joint structure, age, sex, and sport requirements when recommending stretching protocols.

STATIC STRETCHING TECHNIQUES

NECK

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- 2 Flexion and Extension [383](#)

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- 5** Behind-Neck Stretch (Chicken Wing) [385](#)

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Guidelines and Precautions for Static Stretching

- Get into a position that facilitates relaxation.
- Move to the point in the ROM where you experience a sensation of mild discomfort. If performing partner-assisted PNF stretching, communicate clearly with your partner.
- Hold stretches for 15 to 30 seconds.
- Repeat unilateral stretches on both sides.
- Decrease stretch intensity if you experience pain, radiating symptoms, or loss of sensation.
- Use caution when stretching a hypermobile joint.
- Avoid combination movements that involve the spine (e.g., extension and lateral flexion).
- Stabilizing muscles should be active to protect other joints and prevent unwanted movements.

Neck

1 LOOK RIGHT AND LEFT

1. Stand or sit with the head and neck upright.
2. Turn the head to the right using a submaximal concentric muscle action.
3. Turn the head to the left using a submaximal concentric muscle action.

MUSCLE AFFECTED

sternocleidomastoid



2

FLEXION AND EXTENSION

1. Standing or sitting with head and neck upright, flex the neck by tucking the chin toward the chest.
2. If the chin touches the chest, try to touch the chin lower on the chest.
3. Extend the neck by trying to come as close as possible to touching the head to the back.

MUSCLES AFFECTED

sternocleidomastoid, suboccipitals, splenae



Neck flexion



Neck extension

Shoulders and Chest

3 STRAIGHT ARMS BEHIND BACK

1. Standing, place both arms behind the back.
2. Interlock fingers with palms facing each other.
3. Straighten the elbows fully.
4. Slowly raise the arms, keeping the elbows straight.
5. Keep the head upright and neck relaxed.

MUSCLES AFFECTED

anterior deltoid, pectoralis major



4 SEATED LEAN-BACK

1. Sitting with the legs straight and arms extended, place palms on the floor about 12 inches (30 cm) behind the hips.
2. Point the fingers away from the body (backward).
3. Slide the hands backward and lean backward.

MUSCLES AFFECTED

anterior deltoid, pectoralis major



Posterior of Upper Arm

5 BEHIND-NECK STRETCH (CHICKEN WING)

1. Standing or sitting, abduct the right shoulder and flex the elbow.
2. Reach the right hand down toward the left scapula.
3. Grasp the right elbow with the left hand.
4. Pull the elbow behind the head with the left hand to increase shoulder abduction.
5. Repeat on the other side.

MUSCLES AFFECTED

triceps brachii, latissimus dorsi



Upper Back

6 CROSS ARM IN FRONT OF CHEST

1. Stand or sit with the left elbow slightly flexed (15-30 degrees) and the arm across the body (i.e., shoulder in horizontal adduction).
2. Grasp the upper arm just above the elbow, placing the right hand on the posterior side of the upper arm.
3. Pull the left arm across the chest (toward the right) with the right hand.
4. Repeat on the other side.

MUSCLES AFFECTED

posterior deltoid, rhomboids, middle trapezius



7 ARMS STRAIGHT UP ABOVE HEAD (PILLAR)

1. Stand with the arms in front of the torso, fingers interlocked with palms facing out.
2. Slowly straighten the arms above the head with the palms up.
3. Continue to reach upward with the hands and arms.
4. While continuing to reach upward, slowly reach slightly backward.

MUSCLE AFFECTED

latissimus dorsi



Lower Back

8 SPINAL TWIST (PRETZEL)

1. Sitting with the legs straight and the upper body nearly vertical, place the right foot to the left side of the left knee.
2. Place the back of the left elbow on the right side of the right knee, which is now flexed.
3. Place the right palm on the floor 12 to 16 inches (30-41 cm) behind the hips.
4. Push the right knee to the left with the left elbow while turning the shoulders and head to the right as far as possible. Try to look behind the back.
5. Repeat on the other side.

MUSCLES AFFECTED

internal oblique, external oblique, piriformis, erector spinae



9 SEMI-LEG STRADDLE

1. Sit with knees flexed 30 degrees to 50 degrees; let the legs totally relax.
2. Point the knees outward; the sides of the knees may or may not touch the floor.
3. Lean forward from the waist and reach forward with extended arms.

Note: Flexing the knees and relaxing the legs decreases hamstring involvement and increases lower back stretch.

MUSCLE AFFECTED

erector spinae



10 COBRA STRETCH

1. Lie prone on the floor or mat.
2. Place the hands on the floor in a push-up position just below the shoulders with the elbows flexed and held close to the body.
3. Keep the legs extended with the top of the toes or feet resting on the floor.
4. Lift the head and chest while pressing down through the hands to extend the spine (arch the back).
5. Be sure to keep the pelvis and lower body in contact with the floor while looking forward. The result is only a partial extension of the elbows unless the athlete is extremely flexible.
6. Hold the position while breathing slowly and deeply.
7. Slowly lower back to the floor to return to the beginning position.

MUSCLES AFFECTED

rectus abdominis, pectoralis major, iliopsoas, erector spinae



Hips

11 FORWARD LUNGE (FENCER)

1. Standing, take a long step forward with the left leg and flex the left knee until it is directly over the left foot.
2. Keep the left foot flat on the floor.
3. Keep the right knee moderately flexed.
4. Keep the back foot pointed in the same direction as the front foot; it is not necessary to have the heel on the floor.
5. Keep the torso upright and rest the hands on the hips or the front leg (or allow them to hang at the sides).
6. Slowly lower the hips forward and downward.
7. Repeat on the other side.

MUSCLES AFFECTED

iliopsoas, rectus femoris



12 SUPINE KNEE FLEX

1. Lie on the back with legs straight.
2. Flex the right knee and hip, bringing the thigh toward the chest.
3. Place both hands behind the thigh and continue to pull the thigh toward the chest.
4. Repeat on the other side.

MUSCLES AFFECTED

hip extensors (gluteus maximus and hamstrings)



13 FIGURE FOUR STRETCH

1. Lie supine on the floor.
2. Flex and externally rotate the left hip to cross the left ankle over the right thigh.
3. Flex the right hip and pull it toward the chest until a comfortable stretch is felt.
4. Hold and then release.
5. Repeat on the other side.

MUSCLES AFFECTED

gluteus medius, piriformis



Torso

14

SIDE BEND WITH STRAIGHT ARMS

1. Stand with feet shoulder-width apart.
2. Interlace the fingers with the palms away from the torso and facing outward.
3. Reach upward with straight arms.
4. Keeping the arms straight, lean from the waist to the left side.
Do not flex the knees.
5. Repeat on the other side.

MUSCLES AFFECTED

external oblique, latissimus dorsi, serratus anterior



15

SIDE BEND WITH BENT ARM

1. Stand with feet shoulder-width apart.
2. Flex the right elbow and raise the elbow above the head.
3. Reach the right hand down toward the left shoulder.
4. Grasp the right elbow with the left hand.
5. Pull the elbow behind the head.
6. Keeping the elbow flexed, lean from the waist to the left side.
Do not flex the knees.
7. Repeat on the other side.

MUSCLES AFFECTED

external oblique, latissimus dorsi, serratus anterior, triceps brachii



16 LUNGE WITH SPINAL TWIST

1. Assume a lunge position with the right leg forward.
2. Place the left hand on the floor directly parallel to the right foot.
3. Rotate the body, extending the right arm toward the ceiling, and hold.
4. Repeat on the other side.

MUSCLES AFFECTED

erector spinae, semispinalis, hamstrings, gluteus maximus, iliopsoas, quadriceps, obliques, pectoralis major, anterior deltoids



Anterior of Thigh and Hip Flexor

17 SIDE QUADRICEPS STRETCH

1. Lie on the right side with both legs straight.
2. Place the right forearm flat on the floor and the upper arm perpendicular to the floor.
3. Place the left forearm at a 45-degree angle to the torso.
4. Flex the left leg (knee), with the heel of the left foot moving toward the buttocks.
5. Grasp the front of the ankle with the left hand and pull toward the buttocks.
6. Repeat on the other side.

Note: The stretch occurs as a result of knee flexion and hip extension.

MUSCLES AFFECTED
quadriceps, iliopsoas



Posterior of Thigh

18 SITTING TOE TOUCH

1. Sit with the upper body nearly vertical and legs straight.
2. Lean forward using hip flexion and grasp the toes with each hand. Slightly pull the toes toward the upper body and pull the chest toward the legs. If you have limited flexibility, try to grasp the ankles.

MUSCLES AFFECTED

hamstrings, erector spinae, gastrocnemius



19 SEMISTRADDLE

1. Sit with the upper body nearly vertical and legs straight.
2. Place the sole of the right foot on the inner side of the left knee. The outer side of the right leg should be resting on the floor.
3. Lean forward using hip flexion and grasp the toes of the left foot with the left hand. Slightly pull the toes toward the upper body as the chest is also pulled toward the left leg.
4. Repeat on the other side.

MUSCLES AFFECTED

gastrocnemius, hamstrings, erector spinae



Groin

20 STRADDLE (SPREAD EAGLE)

1. Sit with the upper body nearly vertical and legs straight. Abduct the hips, spreading the legs as far as possible.
2. With both hands, grasp the toes of the left foot and pull on the toes slightly while pulling the chest toward the left leg.

3. Repeat toward the center by grasping the right toes with the right hand and the left toes with the left hand. Pull the torso forward and toward the floor.

MUSCLES AFFECTED

gastrocnemius, hamstrings, erector spinae, hip adductors, sartorius



21 BUTTERFLY

1. Sitting with the upper body nearly vertical and legs straight, flex both knees, bringing the soles of the feet together.
2. Pull the feet toward the body.
3. Place the hands on the feet and the elbows on the legs.
4. Pull the torso slightly forward as the elbows push down, causing hip abduction.